

Daily Integral - Hard

$$\int_{-\infty}^{\infty} \left(1 + \frac{x^2 \cos(\pi)}{\Gamma(2)} + \frac{x^4 \cos(2\pi)}{\Gamma(3)} + \frac{x^6 \cos(3\pi)}{\Gamma(4)} + \dots \right) dx$$

This problem was generously submitted by James Chen on <https://dailyintegral.com>.

Solution

First, recognize that $\Gamma(n) = (n-1)!$ for $n \in \mathbb{Z}^+$:

$$\int_{-\infty}^{\infty} \left(1 + x^2 \cos(\pi) + \frac{x^4 \cos(2\pi)}{2!} + \frac{x^6 \cos(3\pi)}{3!} + \dots \right) dx.$$

Now, we can simplify the sum into a series:

$$\int_{-\infty}^{\infty} \sum_{n=0}^{\infty} \frac{\cos(n\pi) x^{2n}}{n!} dx.$$

Since $\cos(n\pi)$ oscillates from -1 and 1 , it simplifies to $(-1)^n$:

$$\begin{aligned} & \int_{-\infty}^{\infty} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} dx \\ &= \int_{-\infty}^{\infty} \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} dx. \end{aligned}$$

Now, recognize that this is a substitution of $-x^2$ to the Maclaurin series of e^u , hence it is the Maclaurin series of e^{-x^2} :

$$\int_{-\infty}^{\infty} e^{-x^2} dx.$$

Now set this integral equal to I :

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Then, I^2 is equal to

$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2} e^{-y^2} dx dy \\ &= \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy. \end{aligned}$$

Now, we can change from rectangular coordinates to polar coordinates, meaning $x^2 + y^2 = r^2$, and $dx dy = r dr d\theta$, with $r \in [0, \infty]$ and $\theta \in [0, 2\pi]$:

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{\infty} e^{-r^2} r dr \\ &= 2\pi \int_0^{\infty} e^{-r^2} r dr. \end{aligned}$$

For the remaining integral, set $u = r^2$, then $1/2 du = r dr$, with $u \in [0, \infty]$:

$$\begin{aligned} I^2 &= \pi \int_0^{\infty} e^{-u} du \\ &= \pi \cdot -e^{-u} \Big|_0^{\infty} \\ &= \pi \left(-\frac{1}{e^{\infty}} + \frac{1}{e^0} \right) \\ &= \pi. \end{aligned}$$

Hence, if $I^2 = \pi$, then the original integral must equal $\sqrt{\pi}$.